

ALPACA AI TRADING HACKATHON

SPY WEEKLY IRON BUTTERFLY

FULLY AUTONOMOUS LIVE-TRADING AGENT

Team AI Velocity Trading

Data-driven entries and exits, executed through Alpaca

Made with GAMMA

A Proven Strategy, Automated

STRATEGY BLUEPRINT

IRON BUTTERFLY



A market-neutral options structure designed to stay centered around SPY through weekly expiration.

BUY PUT

ATM - \$10

SELL PUT

ATM

SELL CALL

ATM

BUY CALL

ATM + \$10

\$10 WING WIDTH

Defined risk boundaries

MARKET-NEUTRAL STRUCTURE

Centered for range-bound conditions

31 MONTHS OF HISTORICAL BACKTESTING

Validated across market regimes

REPEATABLE EXECUTION

Define structure.
Defined risk.

The Intelligence Is in the Timing

It doesn't blindly trade the clock.

How the agent decides when to enter

1 01 SEARCH BEGINS

Monday · 9:35 ET

2 02 POLL SPY MINUTE BY MINUTE

Measure the trailing 20-minute price range

3 03 WAIT FOR VOLATILITY TO SETTLE

Watch for the trailing range to stop making new lows

4 04 ENTER

After 3 minutes with no new range low

No trigger by 10:30 ET → hard timeout → enter

The agent waits for evidence that morning volatility has stabilized before committing.



From Backtest to Live Decision-Making

BACKTEST

Can evaluate the entire completed historical entry window.

Can identify the quietest minute with hindsight.

LIVE AGENT

Only knows what the market has revealed so far.

Looks for evidence volatility has stabilized.

Commits after 3 minutes without a new trailing-range low.

At 10:30 ET, a hard timeout guarantees execution if no earlier trigger occurs.



■ LIVE MARKET FEED // DATA_SYNC_OK

Same strategy. Live-safe decision logic.

Autonomous Risk Management

Once it's in, it keeps watching.

01 WING BREACH

Immediate close

02 +90% MAX-PROFIT TARGET

Automatically lock in gains

03 -80% MAX-LOSS STOP

Automatically close at the loss
threshold

THURSDAY - 3:40 PM ET

MANDATORY FALLBACK EXIT

Close before Friday expiration if no earlier exit condition fires.

The agent doesn't recommend an exit. It executes it.

Built on Alpaca



PROFILE-BASED AUTHENTICATION

No raw API keys in application code

UNATTENDED OPERATION

Runs autonomously after startup confirmation

SINGLE EXECUTION LAYER

Market data and trading through Alpaca

Every Decision Leaves a Trail

Fully auditable by design.

Every step is recorded for review and analysis.

- **DECISION** — Why the agent entered or exited
- **EXPECTED VALUE** — Pre-trade calculation
- **ORDERS & FILLS** — Execution record
- **MARKET STATE** — Price and position monitoring
- **CSV OUTPUT** — Complete post-trade history

 Every decision.
Every EV calculation.
Every fill.
Logged for post-trade analysis.

SYSTEM_LOG // AUDIT_READY

FULLY AUDITABLE

TRADE EXECUTION LOG

DECISION AUDIT

ID	ASSET	SIDE	EXPECTED VALUE	FILL PRICE	STATUS
#8421	SPY 510P	SELL	+1.24%	3.55	EXECUTED
#8422	SPY 500P	BUY	+0.88%	2.10	EXECUTED
#8423	SPY 510C	SELL	+2.15%	3.65	EXECUTED
#8424	SPY 520C	BUY	+0.45%	1.85	EXECUTED
#8425	SPY 510-520-500	IRON BFLY	+1.02%	1.25	EXECUTED

Historical Validation

Backtest | Jan 2024 – Aug 2026 · 100 contracts per leg

+\$303,600

Cumulative Backtested P&L

60.8%

Win Rate

1.23

Sharpe Ratio

97

Total Trades

-\$75,000

Max Drawdown

Jan 2024 – Aug 2026 backtest at 100 contracts per leg, using SPY 1-minute bars and Alpaca OPRA options pricing.
Parameters were tuned on the same period (in-sample); no commissions or slippage modelled.

Backtested results. Historical performance is not indicative of future results.

Autonomous From Entry to Exit

01

OBSERVE

SPY minute-by-minute

02

DECIDE

Detect market stabilization and trigger entry

03

CONSTRUCT

Build defined-risk 4-leg Iron Butterfly

04

EXECUTE

Route trading through Alpaca

05

MONITOR

Continuously evaluate risk and exit conditions

06

EXIT

Automatically close when an exit rule is triggered

07

LOG

Maintain a complete decision and execution audit trail

From market observation through position exit, the agent combines dynamic decision-making, Alpaca execution, autonomous risk management, and transparent logging in a single workflow.

A backtested options strategy transformed into an autonomous live trader.

Alpaca AI Trading Agents Hackathon

Team AI Velocity Trading